

The minimal model program for varieties of general type

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Outline of the talk

1 Introduction

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- 2 The singularities of the MMP

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Introduction

The aim of the Minimal Model Program is to show that if $X \subset \mathbb{P}_{\mathbb{C}}^N$ is a variety with mild singularities, then there is a finite sequence of well understood birational maps (Flips and Divisorial Contractions)

$$X = X_1 \dashrightarrow X_2 \dashrightarrow \dots \dashrightarrow X_n = \bar{X}$$

such that either

- 1) $\bar{X}$ is a minimal model (i.e. $K_{\bar{X}}$ is nef); or
- 2) $\bar{X} \rightarrow Z$ is a Mori fiber space (i.e. $-K_{\bar{X}}$ is ample over Z).

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The singularities of the MMP

- (X, Δ) is a **log pair** if
 - X is a normal variety;
 - $D = \sum \delta_i \Delta_i \in \text{WDiv}_{\mathbb{R}}(X)$ where Δ_i are prime divisors on X and $\delta_i \in \mathbb{R}_{\geq 0}$;
 - $K_X + \Delta$ is $\mathbb{R}$ -Cartier.

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 - $K_X + \Delta$ is $\mathbb{R}$ -Cartier.
- A **log resolution** of (X, Δ) is a proper birational morphism $f : Y \rightarrow X$ such that:
 - Y is smooth;
 - $\text{Ex}(f) \cup f^{-1}(\Delta)$ has simple normal crossings support.

The singularities of the MMP II

- We write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

where Γ and E are effective with no common components,
 $f_*K_Y = K_X$, $f_*\Gamma = \Delta$ and $f_*E = 0$

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- $E - \Gamma = \sum a(F, X, \Delta) \cdot F$ where $a(F, X, \Delta)$ is the discrepancy of (X, Δ) along F .
- We think of the discrepancies as measuring the singularities of (X, Δ) . Bigger discrepancies correspond to milder singularities.
- (X, Δ) is **KLT** (resp. **LC**) if $a(F, X, \Delta) > -1$ (resp. $a(F, X, \Delta) \geq -1$ for all divisors F over X).

The singularities of the MMP III

- (X, Δ) is **terminal** (resp. **canonical** or **PLT**) if
$$a(F, X, \Delta) > 0 \text{ (resp. } a(F, X, \Delta) \geq 0 \text{ or } a(F, X, \Delta) > -1)$$
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- Terminal surfaces are smooth. Canonical surfaces have isolated DuVal singularities.
- If (X, Δ) is PLT and $\lfloor \Delta \rfloor = S$, then $(K_X + \Delta)|_S = K_S + \Delta_S$ where (S, Δ_S) is KLT.

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- Let $[\Delta] = S = \sum S_i$. Then W is a LC center iff $W = S_{i_1} \cap \dots \cap S_{i_k}$. We have that (W, Δ_W) is DLT where

$$(K_X + \Delta)|_W = K_W = \Delta_W.$$

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Base Point Free Theorem

Recall that an $\mathbb{R}$ -Cartier divisor is **nef** if $D \cdot C \geq 0$ for all curves $C \subset X$ and it is **big** if

$$\mathrm{vol}(D) = \frac{1}{m} \lim_{m \rightarrow \infty} h^0(\mathcal{O}_X(\lfloor mD \rfloor)) > 0.$$

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Theorem

Let (X, Δ) be a projective KLT pair, D a nef $\mathbb{R}$ -Cartier divisor such that $aD - (K_X + \Delta)$ is nef and big for some $a > 0$, then D is semiample.

If moreover aD is Cartier, then $|maD|$ is base point free for any $m \geq m_0$ where m_0 depends only on $\dim X$.

Base Point Free Theorem II

- 1 Recall that an $\mathbb{R}$ -divisor D is **semiample** if there is a morphism $f : X \rightarrow Z$ such that $D \sim_{\mathbb{R}} f^*H$ where H is an ample $\mathbb{R}$ -Cartier divisor on Z .

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- 2 The above result also holds in the relative setting i.e. for projective morphisms $\pi : X \rightarrow U$, where we require that D is nef over U and $aD - (K_X + \Delta)$ is nef over U and big over U , etc.

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- 2 The above result also holds in the relative setting i.e. for projective morphisms $\pi : X \rightarrow U$, where we require that D is nef over U and $aD - (K_X + \Delta)$ is nef over U and big over U , etc.
- 3 Corollary: If $K_X + \Delta$ is KLT, nef and Δ is big, then $K_X + \Delta$ is semiample. (Proof. May assume $\Delta = A + B$ with A ample.)

Cone Theorem

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- Let $\overline{NE}(X) = \{\sum a_i C_i \mid a_i \geq 0\} / \equiv$ and $\overline{NE}(X)_{D \geq 0} = \{[C] \in \overline{NE}(X) \mid D \cdot C \geq 0\}$.

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- Let (X, Δ) be a KLT pair. Assume that X is projective (or more generally consider a projective morphism $\pi : X \rightarrow Z$).
- If $K_X + \Delta$ is not nef, then the Cone Theorem gives us precise information about $\overline{NE}(X)_{(K_X + \Delta) < 0}$.

Cone Theorem II

- 1 There are countably many rational curves $C_i \subset X$ such that $0 < -(K_X + \Delta) \cdot C_i \leq 2 \dim X$ and $\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{(K_X + \Delta) \geq 0} + \sum \mathbb{R}_{\geq 0}[C_i]$.

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- 2 If H is ample and $\epsilon > 0$, then
$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{(K_X + \Delta + \epsilon H) \geq 0} + \sum_{\text{finite}} \mathbb{R}_{\geq 0}[C_i].$$

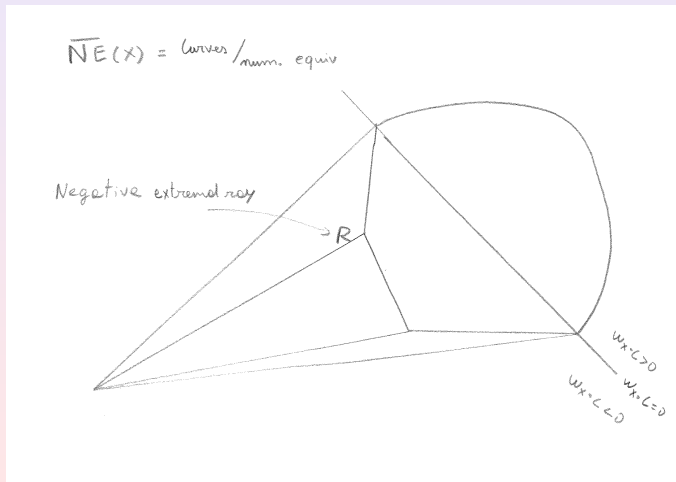
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- 3 If $F \subset \overline{\text{NE}}(X)$ is a $(K_X + \Delta)$ -negative extremal face then there exists a unique $\text{cont}_F : X \rightarrow Z$ such that $(\text{cont}_F)_* \mathcal{O}_X = \mathcal{O}_Z$ and a curve $C \subset X$ is contracted iff $[C] \in F$.

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- 4 If L is a line bundle s. t. $L \cdot C = 0$ for any contracted curve C , then there is a line bundle L_Z on Z with $L = \text{cont}_F^* L_Z$.

The Cone Theorem III



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- If $\dim Z < \dim X$ this is a **Mori fiber space** and we are done.

Running the MMP II

- If $\dim Z = \dim X$ and $\dim \operatorname{Ex}(f) = \dim X - 1$, then f is a **divisorial contraction**. $(Z, f_*\Delta)$ is a $\mathbb{Q}$ -factorial KLT pair and we replace (X, Δ) by $(Z, f_*\Delta)$.

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- If $\dim Z = \dim X$ and $\dim \operatorname{Ex}(f) = \dim X - 1$, then f is a **flipping contraction**. $K_Z + f_*\Delta$ is not $\mathbb{R}$ -Cartier (otherwise $K_X + \Delta = f^*(K_Z + f_*\Delta)$ is not relatively anti-ample). So $(Z, f_*\Delta)$ is not KLT.

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- Since we can not replace (X, Δ) by $(Z, f_*\Delta)$, we hope to replace $f : (X, \Delta) \rightarrow Z$ by the flip $f^+ : (X^+, \Delta^+) \rightarrow Z$.

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- Since we can not replace (X, Δ) by $(Z, f_*\Delta)$, we hope to replace $f : (X, \Delta) \rightarrow Z$ by the flip $f^+ : (X^+, \Delta^+) \rightarrow Z$.
- We require that (X^+, Δ^+) is a $\mathbb{Q}$ -factorial KLT pair isomorphic to X in codimension 1 such that $K_{X^+} + \Delta^+$ is ample over Z (here Δ^+ is the strict transform of Δ).

Running the MMP III

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- If the flip of $f : X \rightarrow Z$ exists, it is unique and if $K_X + \Delta$ is $\mathbb{Q}$ -Cartier, it is given by $\text{Proj}_Z \mathfrak{R}(K_X + \Delta)$ where

$$\mathfrak{R}(K_X + \Delta) = \bigoplus_{m \geq 0} f_* \mathcal{O}_X(m(K_X + \Delta)).$$

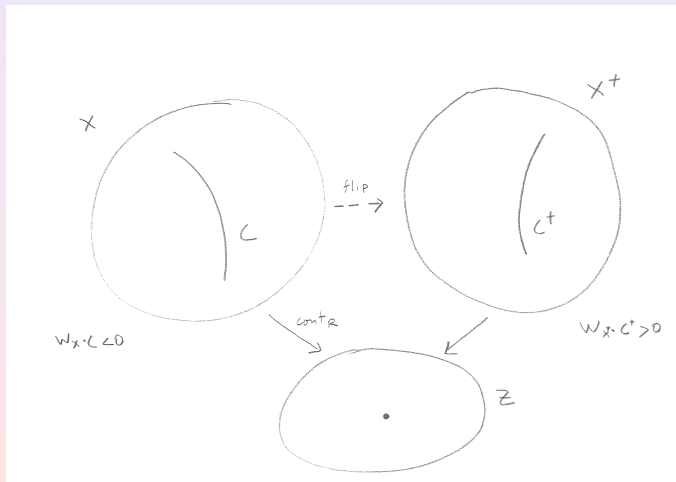
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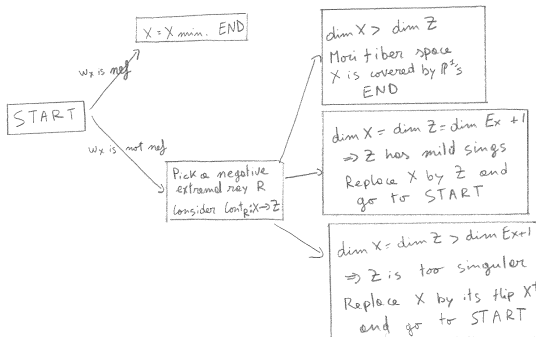
$$\mathfrak{R}(K_X + \Delta) = \bigoplus_{m \geq 0} f_* \mathcal{O}_X(m(K_X + \Delta)).$$

- So to construct the flip we must show that $\mathfrak{R}(K_X + \Delta)$ is finitely generated over Z (i.e. if $Z = \text{Spec} A$, that $R(K_X + \Delta) = \bigoplus_{m \geq 0} H^0(\mathcal{O}_X(m(K_X + \Delta)))$ is a finitely generated A -algebra).

Picture of a flip



MMP flow chart



Running the MMP IV

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- Therefore it suffices to show that there is no infinite sequence of flips.
- The remaining problem is the **Abundance Conjecture**: If (X, Δ) is KLT and $K_X + \Delta$ is nef, then $K_X + \Delta$ is semiample.
- Note that by the Base Point Free Theorem, the Abundance Conjecture is true if $K_X + \Delta$ (or Δ) is big.

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- In dimension 3, termination of flips is easy.
- In these lectures we will show that flips exist in all dimensions and that carefully chosen sequences of flips (those for the MMP with scaling) terminate.
- Termination of flips in dimension 4 is open.

The theorem

Theorem

Let (X, Δ) be a projective KLT pair such that Δ is big (or $K_X + \Delta$ is big, or $K_X + \Delta$ is not pseudo-effective). Then there is a finite sequence of flips and divisorial contractions

$$X = X_1 \dashrightarrow X_2 \dashrightarrow \dots \dashrightarrow X_n = \bar{X}$$

such that $K_{\bar{X}} + \bar{\Delta}$ is nef or $\bar{X} \rightarrow Z$ is a Mori fiber space.

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Corollary

Let (X, Δ) be a projective KLT pair such that Δ is big. Then $R(K_X + \Delta)$ is finitely generated.

Finite generation

Corollary

If X is a smooth projective variety, then $R(K_X)$ is finitely generated.

Proof.

By a result of Fujino and Mori $R(K_X) \cong R(K_Z + \Delta)$ (up to truncations) where (Z, Δ) is a KLT pair and $K_Z + \Delta$ is $\mathbb{Q}$ -Cartier and big. By the Theorem, replacing (Z, Δ) by its minimal model, we may assume that $K_Z + \Delta$ is nef. By the Base Point Free Theorem, $K_Z + \Delta$ is semiample and hence $R(K_Z + \Delta)$ is finitely generated. □

Outline of the talk

- 1 Introduction
- 2 The singularities of the MMP
- 3 Base Point Free and Cone Theorems
- 4 Running the MMP
- 5 Sketch of the proof**
- 6 PI-flips

Sketch of the proof

Let $\dim X = n$. We will break down the proof in to Theorems A, B, C, D, E and then show by induction that

$$(A, B, C, D, E)_{n-1} \Rightarrow A_n \Rightarrow B_n \Rightarrow C_n \Rightarrow D_n \Rightarrow E_n.$$

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Theorem (Theorem A: Existence of Pl-flips)

Pl-flips Exist.

Sketch of the proof II

Theorem (Theorem B: Existence of LTM)

*Let $\pi : X \rightarrow U$ be a projective morphism, (X, Δ) a KLT pair such that Δ is big over U and $K_X + \Delta \sim_{\mathbb{R}, U} D \geq 0$.
Then (X, Δ) has a LTM over U .*

Sketch of the proof II

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Theorem (Theorem C: Non-vanishing)

*Let $\pi : X \rightarrow U$ be a projective morphism, (X, Δ) a KLT pair such that Δ is big over U and $K_X + \Delta$ is pseudo-effective over U .
Then $K_X + \Delta \sim_{\mathbb{R}, U} D \geq 0$.*

Sketch of the proof III

Theorem (Theorem D: Finiteness of Models)

Let $\pi : X \rightarrow U$ be a projective morphism. Let $\mathcal{C} \subset \text{WDiv}_{\mathbb{R}}(X)$ be a rational polytope such that if $K_X + \Delta \in \mathcal{C}$, then Δ is big over U and $K_X + \Delta$ is KLT. Then there are $\phi_i : X \dashrightarrow Y_i$ with $1 \leq i \leq k$ such that

- 1) If $K_X + \Delta \in \mathcal{C} \cap \text{PSEF}(X/U)$ then there is $1 \leq j \leq k$ such that ϕ_j is a LTM of $K_X + \Delta$ over U .
- 2) If $K_X + \Delta \in \mathcal{C}$ and $\phi : X \dashrightarrow Y$ is a LTM of $K_X + \Delta$ over U , then there exists $1 \leq j \leq k$ such that $\xi = \phi_j \circ \phi^{-1} : Y \dashrightarrow Y_j$ is an isomorphism.

Sketch of the proof V

Theorem (Corollary E: Zariski Decomposition)

If $K_X + \Delta$ is KLT and projective, Δ is big and $K_X + \Delta$ is pseudo-effective. Then

1) *(X, Δ) has a LTM and if $K_X + \Delta$ is $\mathbb{Q}$ -Cartier, $R(K_X + \Delta)$ is finitely generated.*

2) *If $V \subset \text{WDiv}_{\mathbb{R}}(X)$ and $\dim V < \infty$, then there exists $\delta > 0$ such that if $\Delta' \in V$, $\|\Delta - \Delta'\| \leq \delta$, then*

$\text{Fix}(K_X + \Delta) \subset \text{Fix}(K_X + \Delta')$.

3) *If $W \subset \text{WDiv}_{\mathbb{R}}(X)$ is the smallest rational affine subspace containing Δ , then there exists $\eta > 0$, $r \in \mathbb{Z}_{>0}$ such that if $\Delta' \in W$, $k(K_X + \Delta')/r$ is integral and $\|\Delta - \Delta'\| \leq \eta$, then $\text{Fix}(k(K_X + \Delta')) \subset \text{Fix}(K_X + \Delta)$.*

Log Terminal Models

Definition

$\phi : X \dashrightarrow Y$ is a LTM (over U) for a KLT pair $K_X + \Delta$ if

- ① ϕ is proper birational map (over U) and ϕ^{-1} contracts no divisors;
- ② $Y \rightarrow U$ is projective;
- ③ Y is normal, and $\mathbb{Q}$ -factorial;
- ④ $K_Y + \phi_*\Delta$ is nef (over U);
- ⑤ $a(F, X, \Delta) < a(F, Y, \phi_*\Delta)$ for any effective divisor $F \subset X$.

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- If $k(K_X + \Delta)$, $k(K_Y + \phi_*\Delta)$ are Cartier, then
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- **Fix** $(K_X + \Delta) = \text{Supp}(E) = \{F \subset \text{Ex}(f)\}$.
- Since Δ is big, the Base Point Free Theorem implies that $K_Y + \phi_*\Delta$ is semiample and hence $R(K_Y + \phi_*\Delta)$ is finitely generated

Log Terminal Models III

- In general the **stable base locus** $\mathbf{B}(D)$ is the intersection of $\text{Supp}(D')$ for all $D \sim_{\mathbb{R}} D' \geq 0$. and the **stable fixed divisor** $\mathbf{Fix}(D)$ is the union of the divisors contained in $\mathbf{B}(D)$.

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- If D is $\mathbb{Q}$ -Cartier $\mathbf{B}(D) = \bigcap_{m>0} \text{Bs}|mD|$.
- The equality $\mathbf{Fix}(K_X + \Delta) = \text{Supp}(E) = \{F \subset \text{Ex}(f)\}$ also holds for $\mathbb{R}$ -divisors.

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- If $k(K_X + \Delta')$ is integral, then so is $k(K_Y + \phi_*\Delta')$. Hence, a fixed multiple of $k(K_Y + \phi_*\Delta')$ is Cartier and so there is an $r \in \mathbb{Z}_{>0}$ such that $rk(K_Y + \phi_*\Delta')$ is base point free.

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- But then $\mathbf{Fix}(rk(K_X + \Delta')) = \text{Supp}(E) = \mathbf{Fix}(K_X + \Delta)$.

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Pl-flips

Definition

Let (X, Δ) be a $\mathbb{Q}$ -factorial PLT pair with $\Delta \in \text{WDiv}_{\mathbb{Q}}(X)$ and $S = \lfloor \Delta \rfloor$. A small birational proper morphism $f : X \rightarrow Z$ is a **pl-flipping contraction** if

- ① $\rho(X/Z) = 1$ and $f_* \mathcal{O}_X = \mathcal{O}_Z$;
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 - ② $-(K_X + \Delta)$ and $-S$ are f -ample
- The **flip** of f is a small birational morphism $f^+ : X^+ \rightarrow Z$ such that X^+ is normal and $\mathbb{Q}$ -factorial, $\rho(X^+/Z) = 1$, $f_*\mathcal{O}_X = \mathcal{O}_Z$ and $K_{X^+} + \Delta^+$ is f^+ -ample.
 - $X^+ = \text{Proj}_Z \bigoplus_{m \geq 0} f_*\mathcal{O}_X(m(K_X + \Delta))$ (the flip is unique and defined locally over Z).

The restricted algebra

Definition

Let $f : (X, \Delta) \rightarrow Z$ be a pl-flip where $S = \lfloor \Delta \rfloor$ and $Z = \text{Spec}(A)$. The **restricted algebra** $R_S(K_X + \Delta)$ is given by

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Idea: Let $s \in \mathbb{Z}_{>0}$, then $R = \bigoplus_{m \geq 0} R_m$ is finitely generated if and only if so is $R^{(s)} = \bigoplus_{m \geq 0} R_{ms}$.

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- As $a(K_X + \Delta) \sim bS'$ for some $a, b \in \mathbb{Z}_{>0}$, $R_S(K_X + \Delta)$ is FG iff $R_S(a(K_X + \Delta))$ is FG iff $R_S(bS')$ is FG iff $R_S(S')$ is FG.

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- Let $g_m \in \mathbb{C}(X)$ such that $g_m \in \text{Ker}(R(S')_m \rightarrow R_S(S')_m)$. Then $(g_m) + mS' = S + B \geq 0$. But then $(g_m/g) + (m-1)S' = B \geq 0$ so $(g_m/g) \in R(S')_{m-1}$ and $\text{Ker}(R(S')_m \rightarrow R_S(S')_m)$ is a principal ideal. QED.

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- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$, then as $K_S + \Delta_S$ is KLT and Δ_S is big over Z , by Thm. E_{n-1} , $R(K_S + \Delta_S)$ is FG.

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- One must show that $\Theta \in \text{WDiv}_{\mathbb{Q}}(S)$ and $|m(K_X + \Delta)|_S = |m(K_S + \Theta_S)| + m(\Delta_S - \Theta_S)$.
- Use extension theorems á la Siu.

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- C defines a contraction $f : X \rightarrow Z$. If $\dim Z < \dim X$, we have a Mori fiber space and we are done.

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- C defines a contraction $f : X \rightarrow Z$. If $\dim Z < \dim X$, we have a Mori fiber space and we are done.
- If $\dim Z = \dim X$, we replace X by the flip or divisorial contraction.

MMP with scaling II

- In this way we either obtain a LTM or a Mori fiber space or a sequence of flips $X_i \dashrightarrow X_{i+1}$ and rational numbers $\lambda_i \geq \lambda_{i+1}$ such that $K_{X_i} + \Delta_i + \lambda_i H_i$ is nef.

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- Note that there is an F over X_i such that $a(F, X_i, \Delta_i) < a(F, X_{i+1}, \Delta_{i+1})$ (any F with center in $\text{Ex}(f)$ will do) and $a(F, X_j, \Delta_j) \leq a(F, X_{j+1}, \Delta_{j+1})$.

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- So X_i is a LTM for $K_X + \Delta + \lambda_i H$ with $0 \leq \lambda_i \leq 1$.
- Note that there is an F over X_i such that $a(F, X_i, \Delta_i) < a(F, X_{i+1}, \Delta_{i+1})$ (any F with center in $\text{Ex}(f)$ will do) and $a(F, X_j, \Delta_j) \leq a(F, X_{j+1}, \Delta_{j+1})$.
- Therefore we have infinitely many LTMs for $K_X + \Delta + \lambda H$ with $0 \leq \lambda \leq 1$. This contradicts Thm. D: Finiteness of LTM.

MMP with scaling III

Corollary

Let (X, Δ) be a projective KLT pair such that Δ is big. Then any sequence of flips and divisorial contractions for the $K_X + \Delta$ MMP with scaling terminates.

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Let (X, Δ) be a projective KLT pair such that Δ is big. Then any sequence of flips and divisorial contractions for the $K_X + \Delta$ MMP with scaling terminates.

- If $K_X + \Delta$ is pseudo-effective, then we obtain a LTM; otherwise we have a Mori fiber space.

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Corollary

Let (X, Δ) be a projective KLT pair such that Δ is big. Then any sequence of flips and divisorial contractions for the $K_X + \Delta$ MMP with scaling terminates.

- If $K_X + \Delta$ is pseudo-effective, then we obtain a LTM; otherwise we have a Mori fiber space.
- Note that the LTM (resp. Mori fiber space) of (X, Δ) is not unique (unless $K_{\bar{X}} + \bar{\Delta}$ is ample). They are related by Flops (Sarkisov links).